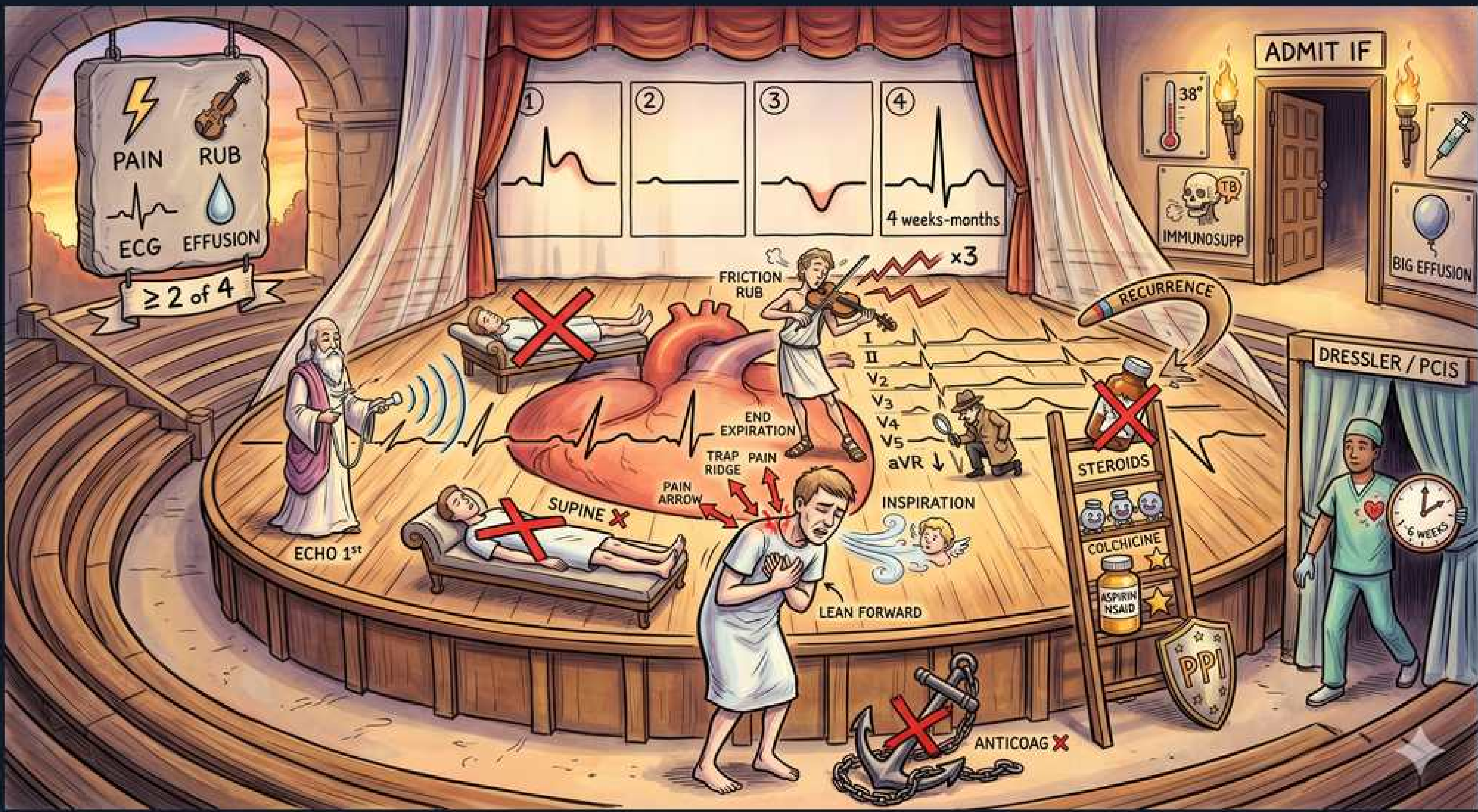
Pericardial Disease — Acute Pericarditis



1. ≥2 of 4 criteria

WHAT YOU SEE

Stone tablet: PAIN, RUB, ECG, EFFUSION with '≥2 of 4'

WHAT IT ENCODES

Acute pericarditis is a CLINICAL diagnosis requiring at least 2 of 4 criteria: (1) characteristic pleuritic chest pain better leaning forward, (2) pericardial friction rub, (3) typical ECG changes (diffuse ST elevation and/or PR depression), and (4) new or worsening pericardial effusion. Supportive findings include elevated CRP/ESR and pericardial inflammation on imaging. Mnemonic: Pain, Rub, ECG, Effusion (PREE) — any 2 seal it.

BOARD TRAP

WRONG to demand all four criteria or a positive troponin before diagnosing pericarditis. Discriminator: only 2 of 4 are required, and troponin is NOT a criterion — an elevated troponin instead suggests myopericarditis (myocardial involvement), which changes prognosis and activity restrictions.

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2. Friction rub

WHAT YOU SEE

Cherub playing a violin (scratchy rub) over the heart

WHAT IT ENCODES

The pericardial friction rub is a scratchy, grating, often triphasic sound (atrial systole, ventricular systole, early diastole) best heard at the left lower sternal border with the patient leaning forward and breath held in end-expiration. It is highly specific (near-pathognomonic) but evanescent — it may come and go, so a single absent exam does not exclude pericarditis. Mnemonic: violin scratch over the heart.

BOARD TRAP

WRONG to exclude pericarditis because no rub is heard on one exam, or to confuse it with a pleural rub. Discriminator: a pericardial rub PERSISTS with breath held (end-expiration) whereas a pleural rub disappears when breathing stops — and the rub is intermittent, so re-auscultate.

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3. Lean forward relieves

WHAT YOU SEE

Patient in gown leaning forward, 'LEAN FORWARD' label

WHAT IT ENCODES

The chest pain of acute pericarditis is sharp/pleuritic and characteristically IMPROVES when sitting up and leaning forward (reduces pressure on the inflamed parietal pericardium) and WORSENS lying flat. It often radiates to the trapezius ridge — a feature quite specific for pericardial irritation via the phrenic nerve. Mnemonic: lean forward = relief; trapezius ache = pericardium.

BOARD TRAP

WRONG to dismiss positional, trapezius-radiating pain as musculoskeletal or anxiety. Discriminator: pain that eases leaning forward, worsens supine/with inspiration, and radiates to the trapezius ridge is classic pericarditis — not the pressure-like, exertional, diaphoresis-associated pain of ischemia.

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4. Supine worsens

WHAT YOU SEE

Patient lying flat on bed with red X, 'SUPINE' label

WHAT IT ENCODES

Pericarditic pain WORSENS lying supine and with inspiration/coughing (pleuritic, positional), which is a key bedside discriminator from myocardial ischemia. The positional and respirophasic nature reflects mechanical irritation of the inflamed pericardium rather than fixed coronary supply-demand mismatch.

BOARD TRAP

WRONG to treat positional, breathing-related chest pain as a STEMI and rush to the cath lab. Discriminator: ischemic pain is typically pressure-like, exertional, and NOT changed by position or respiration, whereas pericarditic pain is sharp, pleuritic, positional (worse supine, better forward).

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5. Diffuse ST elevation

WHAT YOU SEE

Heart with red X plus ECG strips across the stage

WHAT IT ENCODES

Stage 1 ECG shows DIFFUSE (widespread, not territorial) concave-upward ST elevation across multiple leads plus PR depression. Reciprocal changes are absent (except ST depression/PR elevation in aVR and V1). The ST/T ratio in V6 >0.25 favors pericarditis over early repolarization. Mnemonic: pericarditis spreads everywhere and smiles (concave) up.

BOARD TRAP

WRONG to read diffuse ST elevation as a STEMI. Discriminator: STEMI gives REGIONAL, convex ('tombstone') ST elevation in one coronary territory WITH reciprocal depression and often Q waves; pericarditis gives diffuse concave ST elevation with PR depression and NO reciprocal changes (except aVR).

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6. PR depression / aVR

WHAT YOU SEE

ECG leads V2–V6 listed and aVR shown separately

WHAT IT ENCODES

PR-segment depression in most leads is a fairly specific early sign of pericarditis (atrial subepicardial injury). The mirror-image findings in lead aVR (and often V1) are PR ELEVATION and ST depression. Recognizing the aVR reciprocal pattern helps confirm pericarditis. Mnemonic: PR sinks everywhere, aVR is the contrarian that lifts.

BOARD TRAP

WRONG to overlook PR depression or to misread the aVR PR elevation as a separate ischemic event. Discriminator: PR depression with reciprocal PR elevation/ST depression specifically in aVR is a pericarditis signature, not evidence of regional infarction.

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7. ECG evolves over weeks

WHAT YOU SEE

'4 weeks–months' written above ECG panels 1–4

WHAT IT ENCODES

The pericarditis ECG evolves through 4 classic stages over weeks to months: Stage 1 — diffuse ST elevation + PR depression; Stage 2 — ST and PR normalize, T waves flatten; Stage 3 — diffuse T-wave inversions; Stage 4 — return to baseline. Many patients do not show every stage. Mnemonic: up, normalize, invert, resolve.

BOARD TRAP

WRONG to interpret the Stage 3 diffuse T-wave inversions as evolving ischemia/infarction. Discriminator: in pericarditis the T-wave inversions appear AFTER the ST segments have normalized, whereas in ischemia T-wave inversion typically accompanies or follows ST elevation while it is still present, often with territorial Q waves.

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8. NSAIDs first-line

WHAT YOU SEE

A pill bottle of aspirin/NSAID sitting on the medication cart — the first-line anti-inflammatory drug for acute pericarditis.

WHAT IT ENCODES

First-line therapy is a high-dose NSAID — e.g., ibuprofen 600-800 mg every 8 h or aspirin 750-1000 mg every 8 h (aspirin preferred when pericarditis is post-MI), tapered over weeks with a PPI for gastric protection. NSAIDs control pain and inflammation; response also supports the diagnosis. Mnemonic: NSAID + colchicine = the pericarditis duo.

BOARD TRAP

WRONG to use a corticosteroid as the initial anti-inflammatory in routine idiopathic/viral pericarditis. Discriminator: NSAIDs (plus colchicine) are first-line; early steroids are associated with HIGHER recurrence rates and are reserved for specific indications (NSAID contraindication, autoimmune/uremic cause, or refractory disease).

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9. Colchicine add-on

WHAT YOU SEE

A colchicine jar wearing a gold star on the cart — the gold star marks colchicine as the co-first-line partner that halves recurrence.

WHAT IT ENCODES

Colchicine (0.5 mg once or twice daily for ~3 months) is added to NSAIDs as co-first-line: it roughly HALVES the rate of recurrence and improves symptom resolution (ICAP/COPE trials). Main side effect is diarrhea; reduce dose in renal impairment and avoid with strong CYP3A4 inhibitors. Mnemonic: colchicine cuts recurrence in half.

BOARD TRAP

WRONG to omit colchicine and give an NSAID alone. Discriminator: evidence strongly supports NSAID PLUS colchicine — leaving out colchicine forfeits a major reduction in recurrence, which is the most common complication of acute pericarditis.

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10. Avoid steroids early

WHAT YOU SEE

'STEROIDS' bottle with red X

WHAT IT ENCODES

Corticosteroids are AVOIDED as first-line because they increase the risk of recurrent pericarditis (and can foster steroid dependence). They are reserved for NSAID/colchicine failure or contraindication, autoimmune/connective-tissue causes, or uremic pericarditis — and then at low-to-moderate dose with slow taper plus colchicine. Mnemonic: steroids → more recurrences.

BOARD TRAP

WRONG to start prednisone for a typical viral/idiopathic pericarditis to 'calm it down fast.' Discriminator: early steroids paradoxically raise recurrence; the correct first-line is NSAID + colchicine, with steroids only for specific refractory or autoimmune/uremic indications.

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11. Avoid anticoagulants

WHAT YOU SEE

Anchor labeled 'ANTICOAG' with red X

WHAT IT ENCODES

Anticoagulation should generally be AVOIDED in acute pericarditis because it can convert a serous effusion into a hemorrhagic one and precipitate tamponade. If anticoagulation is otherwise mandatory (e.g., mechanical valve), it is used with caution and close monitoring. Mnemonic: inflamed pericardium + blood thinner = bleeding into the sac.

BOARD TRAP

WRONG to anticoagulate a pericarditis patient who also has, say, atrial fibrillation, without weighing tamponade risk. Discriminator: in active pericarditis anticoagulants risk hemorrhagic effusion/tamponade, so they are avoided or used only when truly required with monitoring — unlike in ACS where anticoagulation is standard.

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12. Viral/idiopathic #1

WHAT YOU SEE

The whole scene is staged in a theater/auditorium — the 'stage' setting cues that viral/idiopathic is the common, everyday backdrop of acute pericarditis.

WHAT IT ENCODES

The most common cause of acute pericarditis in developed countries is VIRAL or idiopathic (often Coxsackievirus B / enteroviruses), and it is usually self-limited, resolving with NSAID + colchicine. Extensive etiologic work-up is unnecessary in low-risk, typical presentations; pursue causes (TB, autoimmune, malignancy, uremia) only when high-risk features are present. Mnemonic: most pericarditis is 'just a virus.'

BOARD TRAP

WRONG to launch a broad infectious/autoimmune/malignancy work-up in a low-risk, typical case. Discriminator: in the absence of high-risk features, viral/idiopathic is assumed and no exhaustive testing is needed — targeted work-up is reserved for high-risk or atypical presentations.

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13. Admit criteria

WHAT YOU SEE

Doorway 'ADMIT IF': fever 38°, immunosuppressed, big effusion

WHAT IT ENCODES

High-risk features mandating hospital admission and a search for a specific cause: fever >38°C, subacute onset, large effusion (>20 mm) or tamponade, immunosuppression, oral anticoagulant use, elevated troponin (myopericarditis), and failure to respond to NSAIDs after 7 days. Mnemonic: Fever, big Effusion, Anticoag, Immunosuppression, Trauma/troponin, NSAID failure = admit.

BOARD TRAP

WRONG to discharge a pericarditis patient with fever >38°C, a large effusion, or a positive troponin as 'just viral.' Discriminator: these high-risk markers predict complications (tamponade, specific/serious etiology, myopericarditis) and warrant admission and work-up rather than outpatient management.

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14. Dressler / late

WHAT YOU SEE

Surgeon by door 'DRESSLER / PCIS' with clock '2 weeks'

WHAT IT ENCODES

Post-cardiac-injury syndromes include Dressler syndrome — an autoimmune/immune-mediated pericarditis appearing WEEKS (typically 2-6 weeks) after MI or cardiac surgery, with fever, pleuritic pain, effusion, and elevated inflammatory markers. Distinguish from early peri-infarction (epistenocardiac) pericarditis, which occurs within days of MI. Treat with NSAID/aspirin + colchicine; avoid steroids early. Mnemonic: Dressler = delayed, immune.

BOARD TRAP

WRONG to confuse delayed Dressler syndrome with acute reinfarction or with the early peri-infarct pericarditis seen within the first few days. Discriminator: Dressler appears WEEKS after the injury and is immune-mediated with constitutional symptoms, whereas early post-MI pericarditis is within days, and reinfarction shows new ischemic ECG/biomarker territorial changes.
